

Project Master Plan: Agri-Waste Biomass Valuation Engine (AVBE)

AI-Powered Crop Residue Management and Marketplace Optimization

Competition Theme: Environmental Sustainability

Document Status: Confidential Team Masterplan / Technical Specification

1. The Problem

Every year, agricultural regions across India face a massive environmental and economic crisis during the post-harvest transition window. The burning of crop residue, particularly paddy straw (stubble) in Northern India, releases catastrophic amounts of greenhouse gases (\$CO_2\$, \$CH_4\$) and particulate matter (\$PM_{2.5}\$, \$PM_{10}\$), severely impacting air quality across the subcontinent. This problem persists due to three systemic flaws:

- **Critically Narrow Seasonal Windows:** Farmers have a highly constrained timeline (often less than 14–21 days) to clear their fields after harvesting one crop and preparing the soil for the next planting cycle. Manual collection or waiting for slow traditional decomposition is impossible within this window.
- **Highly Fragmented Biomass Supply Chain:** Industrial buyers (such as biofuel processing plants, paper mills, and sustainable packaging manufacturers) are eager to buy crop residue. However, they have no visibility into which fields have available stubble, the total volume distributed across smallholder farms, or how to coordinate large-scale pickup logistics efficiently.
- **Complete Lack of Transparent Valuation:** Smallholder farmers view crop residue entirely as a disposal liability or waste product rather than a monetizable asset. Because there is no standardized, accessible way to measure the economic value of their specific field's residue, burning remains the fastest, zero-cost choice for them.

2. Current Solutions & Their Limitations

Existing approaches to tackle stubble burning focus on heavy machinery or post-facto tracking, both of which miss the core socio-economic problem:

- **Subsidized Mechanical Management (In-Situ/Ex-Situ):** Governments provide subsidies for heavy industrial balers and happy-seeders. However, the upfront operational costs, tractor fuel requirements, and machinery maintenance remain prohibitively high for low-income, smallholder farmers. The machinery also suffers from a low utilization rate outside the harvest weeks.
- **Manual Biomass Procurement:** Commercial bio-energy aggregators rely on physical field scouting to spot available residue, negotiate pricing with individual farmers, and

organize logistics. This manual process is unscalable, highly subjective, slow, and completely fails to map thousands of scattered rural plots in a short timeframe.

- **Reactive Satellite Thermal Tracking:** Space and environmental agencies use satellite thermal anomalies (such as NASA FIRMS) to detect active farm fires. While highly accurate for statistics, this is entirely reactive—it reports on environmental destruction *after* it occurs, rather than providing a proactive economic alternative that prevents the fire from being lit in the first place.

3. The Solution

The proposed solution is a pure software, cloud-based platform that transforms crop residue from an environmental hazard into a transparently priced commodity, requiring zero custom hardware. By combining localized smartphone imagery with macro-level satellite observations, the engine digitizes the supply chain to eliminate field burning entirely.

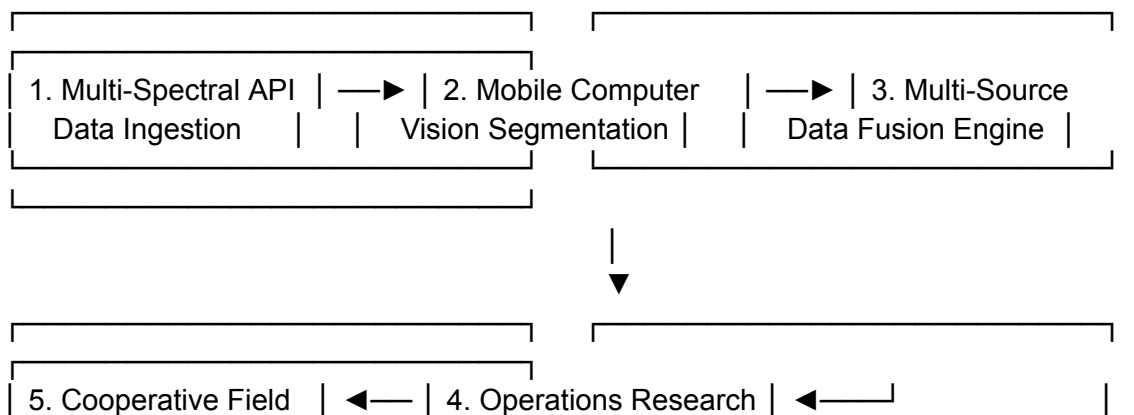
The system operates through three primary software capabilities:

- **Hyper-Local Biomass Volume Estimation:** Computer vision models analyze standard smartphone photos uploaded by farmers to instantly calculate the approximate tonnage of crop residue present on a specific agricultural plot.
- **Dynamic, Quality-Based Pricing Models:** Image analysis combined with local weather data estimates moisture, degradation, and purity levels to determine an objective, fair-market valuation for the crop residue based on real-time industrial demand.
- **Multi-Agent Logistics Optimization:** An operations research backend aggregates thousands of fragmented smallholder plots into optimized logistical clusters, automatically generating highly efficient multi-stop collection routes to minimize transit costs for industrial biomass buyers.

4. How to Make It Happen

To take this platform from architecture to a field-validated marketplace, execution relies entirely on a modular cloud-based software pipeline and a targeted community onboarding framework.

A. End-to-End Technical Workflow



1. **Multi-Spectral Data Ingestion:** Establish continuous data pipelines fetching open-source satellite imagery (Sentinel-2/Landsat) covering primary agricultural belts to track crop maturity curves and predict exact harvest dates.
2. **Mobile Computer Vision Segmentation:** Deploy a deep learning image segmentation model that analyzes a farmer's uploaded smartphone image to distinguish standing stubble or loose residue from bare soil and weeds.
3. **Multi-Source Valuation Fusion:** Combine pixel-level mobile features with macro satellite vegetative indices to compute the total available biomass weight and cross-reference it with current industrial market rates to generate a live quote.
4. **Operations Research Logistics Routing:** Run specialized vehicle routing problem (VRP) algorithms to compile individual farm listings into unified pickup batches, calculating the shortest path for baling trucks.
5. **Cooperative Field Validation Pilot:** Partner with a regional rural cooperative or Farmer Producer Organization (FPO) to test the app with a network of local fields, measuring the exact variance between the AI's volume predictions and the actual weighed truck tonnage at the processing plant.

B. Technical Architecture Specification

Mobile Client Layer (The Farmer Interface)

The frontend application is built using a cross-platform framework (such as Flutter) to remain lightweight, running efficiently on low-tier, budget devices. It operates primarily in regional Indian languages and features a guided camera UI. It forces the user to snap pictures at specific angles while capturing the phone's internal compass orientation and precise GPS coordinates to verify that the photo corresponds directly with registered land plots.

Cloud Inference Pipeline (The Intelligence Core)

The backend processing system consists of three modular neural and mathematical layers:

- **Computer Vision Segmentation Layer (U-Net/MobileNet):** The uploaded mobile image is passed into a modified, highly efficient **U-Net** architecture with a **MobileNetV3** backbone for semantic segmentation. The network classifies pixels into three distinct masks: **Residue_Density**, **Soil_Background**, and **Weed/Foreign_Matter**.

$$\text{Density Ratio} = \frac{\sum \text{Pixels}_{\text{Residue}}}{\sum \text{Pixels}_{\text{Total}}}$$

- **Geospatial Macro Analytics Layer:** The platform queries the European Space Agency's (ESA) **Sentinel-2 API** using the photo's GPS coordinates. It extracts multi-spectral bands to calculate the **NDVI** (Normalized Difference Vegetation Index) and **NDRE** (Normalized Difference Red Edge Index) of the specific land registry polygon over a 30-day timeline. By observing the sharp drop in NDVI (indicating a successful harvest),

the engine validates the plot area and calculates baseline plant density before processing the mobile camera's micro-density ratio.

- **Operations Research Optimization Engine:** Once hundreds of fields are listed on the dashboard, the backend models the region as a geographic coordinate network. It runs a **Mixed-Integer Linear Programming (MILP)** algorithm to solve a Capacitated Vehicle Routing Problem (CVRP). The objective function minimizes fuel consumption and time metrics for the buyer's fleet:

$$\text{Minimize } \sum_i \sum_j C_{ij} X_{ij}$$

Where C_{ij} represents the transport cost between farm i and farm j , and X_{ij} is the binary routing decision matrix.

Technical Note on Environmental and Light Variability

Outdoor rural lighting varies significantly depending on the time of day, dust levels, and cloud coverage, which can skew standard color-based pixel estimation. To maintain consistency, the software pipeline automatically converts raw RGB inputs into the **HSV** (Hue, Saturation, Value) and **Lab color space** configurations. This allows the segmentation layer to evaluate physical fiber texture and structural line distribution rather than relying purely on color tones, keeping the biomass estimation error margin under $\pm 7\%$.

C. Data Strategy

Training robust computer vision and remote sensing models for diverse agricultural landscapes requires a meticulous approach to data gathering and structural alignment.

- **Cross-Modal Ground-Truth Creation:** To calibrate our volume estimation algorithms, we will construct a parallel dataset. We will gather field photographs from diverse angles, pair them with corresponding local satellite coordinates, and match them exactly against the ultimate, verified physical weight of the harvested biomass measured at weighbridges.
- **Synthetic Lighting & Lens Augmentation:** Rural smartphone photos often feature low contrast, dirty camera lenses, or overexposed mid-day sunlight. We will apply heavy data augmentation—including random contrast adjustments, simulated lens smudges, and shadow masking—to the training dataset to ensure the U-Net model functions reliably under real-world conditions.
- **Open-Source Agricultural Benchmarking:** The remote sensing pipeline leverages historical crop classification maps from public frameworks, such as the Google Earth Engine (GEE) repository, to cross-check local soil types and historical regional crop yields over the last 5 years.

5. Development Roadmap

Month 1-3: Mobile CV Models & API Pipelines

- └─ Build Sentinel-2/GEE data ingestion scripts
- └─ Train U-Net/MobileNet on annotated stubble imagery

Month 4-8: Valuation Logic & Core Marketplace Integration

- └─ Develop NDVI-to-biomass weight mapping models
- └─ Build CVRP logistics routing optimization algorithm
- └─ Design localized, multi-lingual Android app interface

Month 9-12: Simulated Testing & Harvest Field Pilot

- └─ Conduct stress testing with simulated farm plot datasets
- └─ Launch field pilot with a local FPO during harvest season

- **Months 1–3 (Model Engineering & Satellite Ingestion):** Develop the data ingestion architecture for satellite APIs. Collect initial crop residue datasets and train the core U-Net computer vision model to achieve high-accuracy residue segmentation across varying outdoor illumination levels.
- **Months 4–8 (Valuation Engine & Mobile App Development):** Write the data fusion algorithms linking local density ratios with satellite index metrics to estimate total tonnage. Implement the MILP operations research algorithm for vehicle routing. Build the localized, intuitive Android application shell and coordinate with backend server architectures.
- **Months 9–12 (System Evaluation & Field Pilot Deployment):** Run systemic simulation tests mapping thousands of synthetic farm listings to evaluate logistics scalability. Launch a live field pilot in collaboration with an FPO or agricultural cooperative during the primary post-monsoon harvest season, directly measuring emissions saved and farmer revenue generated.